WEATHER PREDICTION USING MACHINE LEARNING ALGORITHMS

Submitted To:

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1. ABSTRACT

Weather prediction plays an important role in agriculture, transportation, and daily activities. Machine learning techniques can analyze historical weather data and predict future weather conditions. In this project, the Seattle Weather dataset is used to build and compare different machine learning models. The algorithms used are K-Nearest Neighbors (KNN), Support Vector Machine (SVM), Gradient Boosting, and Extreme Gradient Boosting (XGBoost). The performance of each model is evaluated using accuracy, precision, recall, F1-score, and confusion matrix.

2. OBJECTIVE

The objectives of this project are:

- To analyze the Seattle weather dataset.
- To preprocess and clean the data.
- To implement multiple machine learning algorithms.
- To compare the performance of different models.
- To identify the best model for weather prediction.

3. TOOLS AND TECHNOLOGIES USED

- Python
- Jupyter Notebook
- Pandas
- NumPy
- Scikit-learn

- XGBoost
- Matplotlib
- Seaborn

4. DATASET DESCRIPTION

Dataset Name: Seattle Weather Dataset

Number of Records: 1461

Attributes:

- Date
- Precipitation
- Maximum Temperature
- Minimum Temperature
- Wind
- Weather Condition

Target Variable:

- Weather

```
import pandas as pd

df = pd.read_csv("seattle-weather.xls")

df.head()
df.info()
df.isnull().sum()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1461 entries, 0 to 1460
Data columns (total 6 columns):
 #   Column         Non-Null Count  Dtype
---  ------         --------------  -----
 0   date           1461 non-null   object
 1   precipitation  1461 non-null   float64
 2   temp_max       1461 non-null   float64
 3   temp_min       1461 non-null   float64
 4   wind           1461 non-null   float64
 5   weather        1461 non-null   object
dtypes: float64(4), object(2)
memory usage: 68.6+ KB
```

```
date             0
precipitation    0
temp_max         0
temp_min         0
wind             0
weather          0
dtype: int64
```

5. DATA PREPROCESSING

The following preprocessing steps were performed:

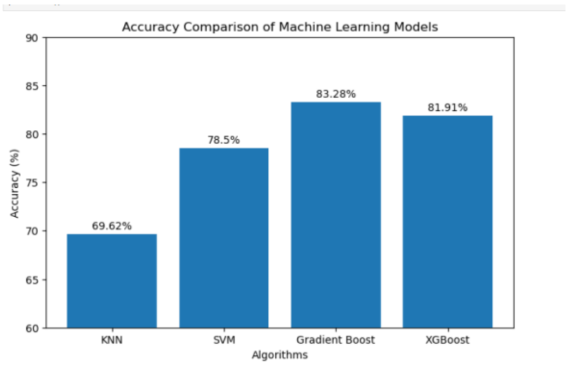
- Removed the date column.
- Converted weather labels into numerical values using Label Encoding.
- Applied feature scaling.
- Split the dataset into training and testing datasets.

6. MACHINE LEARNING ALGORITHMS USED

1. K-Nearest Neighbors (KNN)
2. Support Vector Machine (SVM)
3. Gradient Boosting
4. XGBoost

7. RESULTS

ALGORITHM	ACCURACY
KNN	69.62%
SVM	78.5%
Gradient Booster	83.28%
XGBoost	81.91%



8. CLASSIFICATION REPORT

The classification report evaluates the model using:

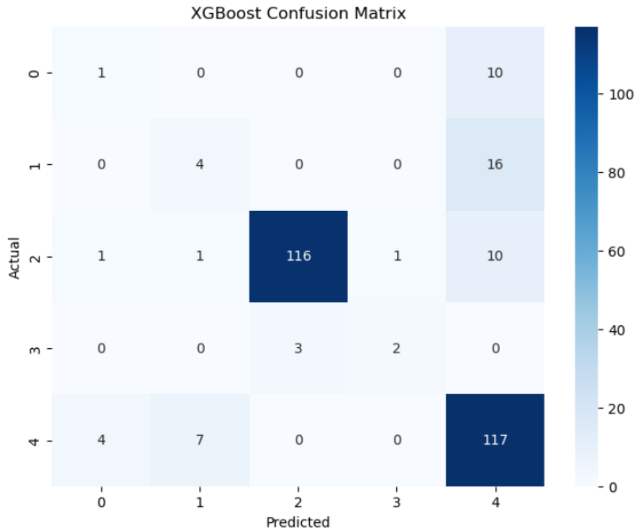
- Precision

- Recall
- F1-Score

```
KNN Accuracy:
0.6962457337883959

SVM Accuracy:
0.7849829351535836

Gradient Boost Accuracy:
0.8327645051194539

XGBoost Accuracy:
0.8191126279863481

Classification Report:

              precision    recall  f1-score   support

           0       0.17      0.09      0.12        11
           1       0.33      0.20      0.25        20
           2       0.97      0.90      0.94       129
           3       0.67      0.40      0.50         5
           4       0.76      0.91      0.83       128

    accuracy                           0.82       293
   macro avg       0.58      0.50      0.53       293
weighted avg       0.80      0.82      0.81       293
```

9. CONFUSION MATRIX

The confusion matrix helps visualize the prediction performance of the model.



10. PERFORMANCE ANALYSIS

Among all the algorithms, Gradient Boosting achieved the highest accuracy of 83.28%. The model performed particularly well for rain and sunny weather classes. The lower performance for drizzle and snow classes is due to the smaller number of samples in the dataset.

11. CONCLUSION

This project successfully implemented multiple machine learning algorithms for weather prediction. The Gradient Boosting model achieved the best performance with an accuracy of 83.28%. The project demonstrates that ensemble learning methods can provide effective weather prediction results.

12. FUTURE SCOPE

- Use larger weather datasets.
- Include humidity and pressure features.
- Apply deep learning models.
- Develop a real-time weather prediction system.